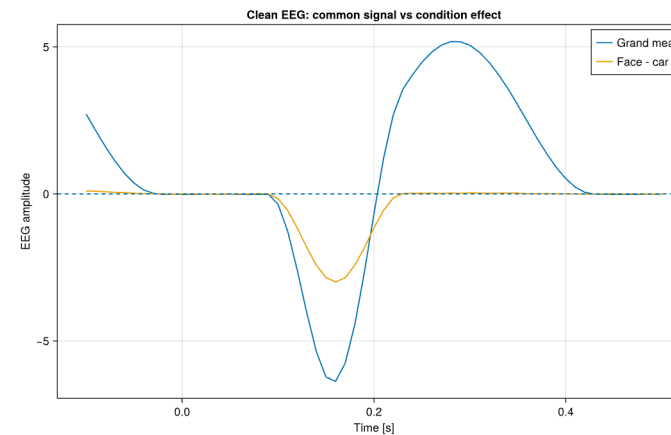
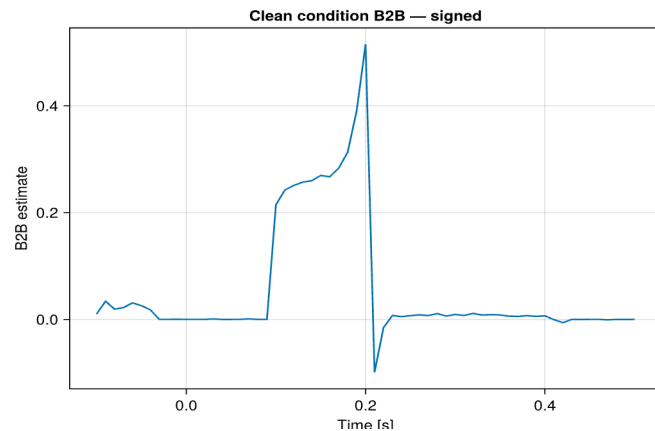
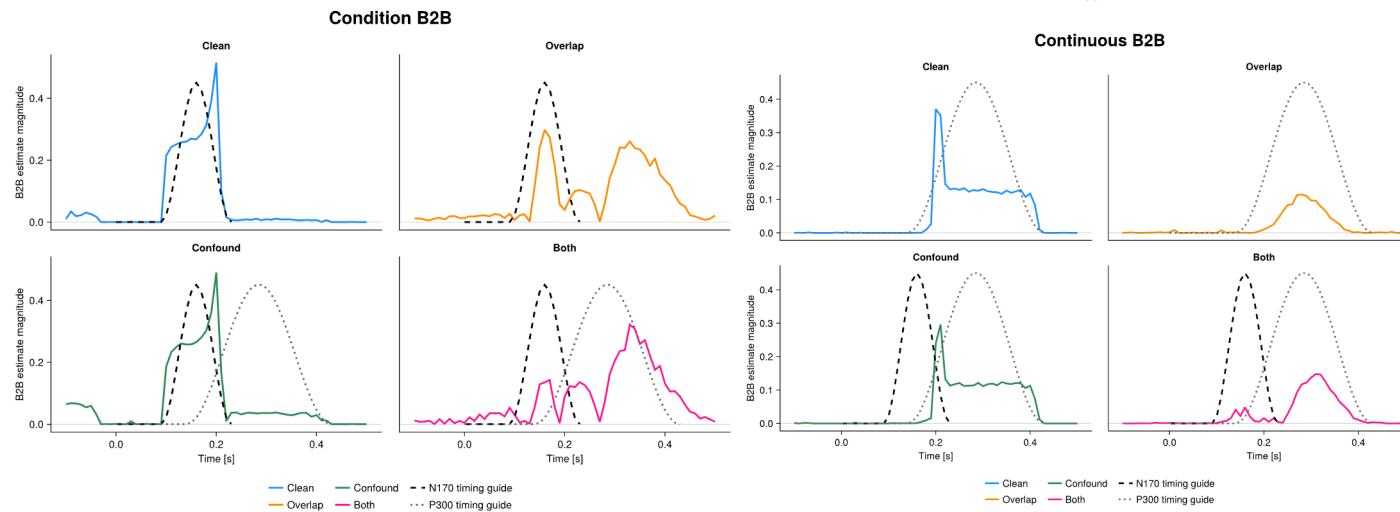


## Shared ERP activity alters B2B recovery: N170–P300 cancellation around 200 ms

Plain B2B on normal epoched data      $\beta_{0\_n170} = 5.0$ ,  $\beta_{0\_p300} = 5.0$ ,      $\beta_{\text{condition}} = 3.0$ ,  $\beta_{\text{continuous}} = 2.0$

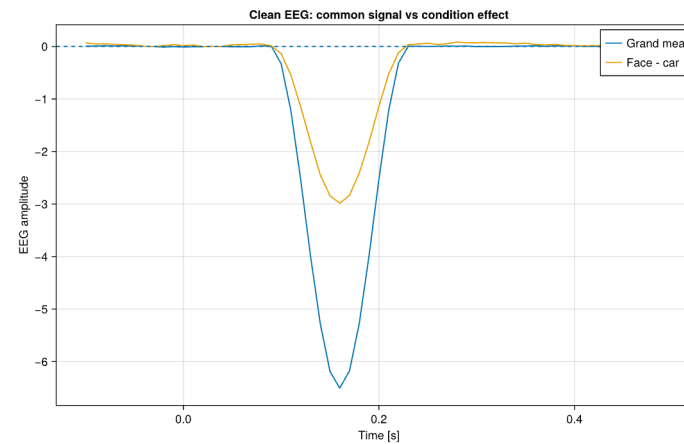
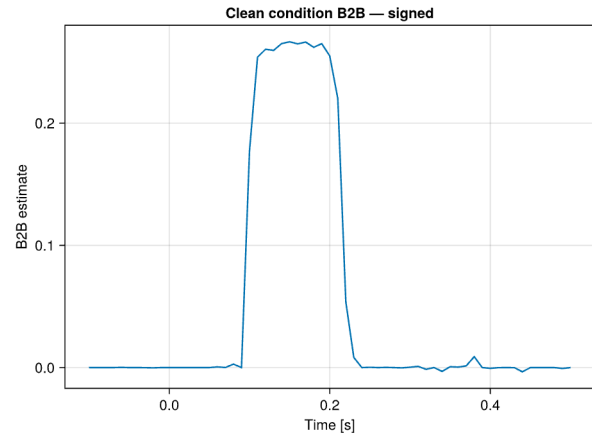


- Blue line: grand-average EEG across all trials.
- Orange line: Face – Car difference
- A sharp B2B spike appears at ~200 ms.
- The spike coincides with cancellation of the shared N170 and P300 components.



- **Setting the P300 intercept to zero removes this spike.**

## Removing the shared P300 component removes the B2B spike



$\beta_{0\_n170} = 5.0,$   
 $\beta_{0\_p300} = 0.0,$   
 $\beta_{\text{condition}} = 3.0,$   
 $\beta_{\text{continuous}} = 1.0/3.0$

